DEREJE D. JIMA, MSC, PHD

Senior Bioinformatics Scientist | Computational Genomics | Multi-omics | Translational Research

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EDUCATION

PhD, Bioinformatics
North Carolina State University
2022 – 2026

MS in Bioinformatics
Wageningen University
2003 – 2005

BS in Plant Science
Haramaya University
1996 – 2000

PROFESSIONAL PROFILE

- Senior bioinformatics scientist with 16+ years of experience in genomics, epigenomics, and multi-omics data analysis across academic, translational, and collaborative research environments. Expertise in next-generation sequencing, statistical modeling, and development of scalable, reproducible analysis pipelines for complex biological datasets.
- Demonstrated impact in integrating complex biological data to generate actionable scientific insights across cancer genomics, immunology, infectious disease, and environmental health studies, with strong emphasis on study design, analytical rigor, and scientific interpretation.
- Proven leader in cross-functional bioinformatics support, partnering with 50+ investigators across three universities to design sequencing studies, develop analytical workflows, produce high-quality reports, and translate data into defensible scientific conclusions.

SELECTED IMPACT

75+ peer-reviewed publications
Extensive publication record across genomics, epigenomics, and biomedical data science

Cross-institution collaboration
Bioinformatics support and consultation for investigators across multiple universities and research centers

Broad omics expertise
Sequencing, methylation, expression, and integrative analysis across diverse study designs

EXPERIENCE

Senior Research Scholar
North Carolina State University
Nov 2023 – Present Raleigh, NC

- Promoted from Research Scholar in recognition of sustained leadership in bioinformatics, multi-omics analysis, and cross-institutional scientific collaboration
- Lead bioinformatics strategy as CHHE Liaison, partnering with 50+ investigators across three universities to support sequencing study design, multi-omics analysis, and interpretation of high-dimensional datasets
- Serve as co-investigator on multiple NIH-funded projects (R01, R21, and CHHE Center grants), contributing to proposal development, analytical planning, and leadership of bioinformatics components in funded research
- Produce and review analytical plans, scientific reports, and technical summaries that support collaborative research execution and publication

Research Scholar
North Carolina State University
Sep 2015 – Oct 2023 Raleigh, NC

- Developed scalable, reproducible pipelines for genome-wide DNA methylation and RNA-seq analysis, enabling integration of large-scale multi-omics datasets
- Led end-to-end bioinformatics analyses across toxicology, cancer genomics, immunology, and environmental health studies using high-throughput sequencing and complementary molecular data
- Supported multi-institutional collaborations and large cohort projects through workflow development, quality-focused analysis, and integrative interpretation of complex omics datasets
- Contributed to grant proposals, study protocols, and data analysis plans, strengthening the bioinformatics design of funded translational research
- Refined and applied computational workflows for RNA-seq, DNA methylation, and other high-dimensional datasets to improve scalability, consistency, and biological interpretability
- Collaborated closely with interdisciplinary investigators to translate analytical results into publications, presentations, and actionable scientific conclusions

CNTS, Senior Bioinformatician

Walter Reed Army Institute of Research

Jun 2014 – Sep 2015 | Silver Spring, MD

- Conducted bioinformatics research in infectious disease, data integration, and application development for pathogen-focused sequencing studies
- Redesigned a microbial pathogen discovery workflow and initiated a 16S metagenomics pipeline to improve analytical capability for surveillance-oriented research
- Supported surveillance- and vaccine-related research through bioinformatics analysis, data integration, and scientific interpretation
- Contributed genome assembly and downstream sequence analysis for *Candidatus Rickettsia asemboensis* strain NMRCii
- Held U.S. Government Public Trust security clearance

Bioinformatician I

Duke University Medical Center

Mar 2007 – Jun 2014 | Durham, NC

- Contributed to cancer genomics and B-cell biology research through computational analysis, scientific reporting, and manuscript development
- Designed and implemented analytical frameworks for sequencing, gene expression, and epigenetic microarray studies
- Analyzed biological data generated using diverse high-throughput platforms and translated results into publication-ready scientific outputs
- Built, configured, and administered a Linux compute cluster with 16 Supermicro servers to support high-throughput bioinformatics workloads

Research Associate

North Carolina State University

Sep 2005 – Mar 2007 | Raleigh, NC

- Conducted innate immunity research focused on natural killer cells
- Performed independent research, expression microarray analysis, and molecular biology wet-lab work
- Managed laboratory operations, equipment, and regulated biological materials

Visiting Research Scholar

Oklahoma State University

Mar 2005 – Sep 2005 | Stillwater, OK

- Worked on database integration, parsing, Perl scripting, and MySQL-based data mining

Assistant Lecturer

Ambo University

Dec 2000 – Aug 2003 | Ambo, Ethiopia

- Taught university-level courses and conducted independent academic research

TECHNICAL PROFILE

Languages
Python, R/Bioconductor, Perl, Bash, SQL, NoSQL, HTML, CSS, JSON

Platforms & Tools
Linux, Git, PostgreSQL, MySQL, SQLite, Redmine, WordPress, Bootstrap, LaTeX

Data Modalities
RNA-seq, Methyl-seq, ChIP-seq, FAIRE-seq, methylation arrays, exome sequencing, whole-genome sequencing, Tempo-seq

HONORS

PRIDE OF THE WOLFPACK
Recognition for contributions to North Carolina State University (March 2026)

CHHE Outstanding Member Award
North Carolina State University, 2019

Netherlands Fellowship Program
Scholarship support for MS study

Degree with Distinction
Haramaya University

AFFILIATIONS

Memberships
Member, Center for Human Health and the Environment (CHHE), North Carolina State University; Member, Bioinformatics Research Center (BRC), North Carolina State University; Affiliate Member, University of North Carolina at Chapel Hill (UNC), Chapel Hill, NC (2016–Present); Member, International Society of Developmental and Comparative Immunology (2005–2007)

Professional Service
Former editorial board member, Austin Medical Sciences Journal; Peer reviewer for genomics and biomedical journals (3+ manuscripts annually)

KEY STRENGTHS

Bioinformatics | Epigenomics | DNA Methylation | RNA-seq | WGS/WES | Multi-omics | Pipeline Development | Statistical Genomics | Study Design | Linux/HPC

Python | R/Bioconductor | Perl | Bash | SQL | NoSQL | Git | LaTeX

SELECTED PUBLICATIONS

- D. D. Jima, D. A. Skaar, A. Planchart, A. Motsinger-Reif, S. E. Cevik, and S. S. Park, "Genomic map of candidate human imprint control regions: The imprintome," *Epigenetics*, vol. 17, no. 13, pp. 1920–1943, 2022.
- J. Zhang et al., "The genomic landscape of mantle cell lymphoma is related to the epigenetically determined chromatin state of normal b cells," *Blood*, vol. 123, no. 19, pp. 2988–2996, 2014.
- J. Zhang, V. Grubor, C. L. Love, A. Banerjee, K. L. Richards, and P. A. Mieczkowski, "Genetic heterogeneity of diffuse large b-cell lymphoma," *Proceedings of the National Academy of Sciences*, vol. 110, no. 4, pp. 1398–1403, 2013.
- C. Love et al., "The genetic landscape of mutations in burkitt lymphoma," *Nature Genetics*, vol. 44, no. 12, pp. 1321–1325, 2012.
- **D. D. Jima**, J. Zhang, C. Jacobs, K. L. Richards, C. H. Dunphy, and W. W. L. Choi, "Deep sequencing of the small rna transcriptome of normal and malignant human b cells identifies hundreds of novel micrornas," *Blood*, vol. 116, no. 23, e118–e127, 2010.
- J. Zhang et al., "Patterns of microrna expression characterize stages of human b-cell differentiation," *Blood*, vol. 113, no. 19, pp. 4586–4594, 2009.

LINKS TO FULL PUBLICATIONS

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